

CSL-220: Database Management Systems Lab

LAB # 3

SINGLE ROW FUNCTIONS

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SINGLE ROW FUNCTIONS

Single-row functions in SQL operate on single rows to produce a result based on each row. Some of them are enlisted below.

- Numeric Functions
- Character/String Functions
- Date Functions
- Conversion Functions
- General/Miscellaneous Functions

NUMERIC FUNCTIONS

Numeric functions in SQL operate on numeric values and return a numeric result.

1. **ABS(x)**: Returns the absolute value of x.
2. **CEIL(x) / CEILING(x)**: Returns the smallest integer greater than or equal to x.
3. **FLOOR(x)**: Returns the largest integer less than or equal to x.
4. **ROUND(x,n)**: Rounds x to n decimal places (default is 0).
5. **SQRT(x)**: finding square root of a number.

CHARACTER/STRING FUNCTIONS

Character functions accept character inputs and can return either characters or number values as output.

Example:

- **UPPER(string)**: Converts all characters in a string to uppercase.
- **LOWER(string)**: Converts all characters in a string to lowercase.
- **LENGTH(string)**: Returns the length of the string.
- **CONCAT(string1, string2)**: Concatenates two strings.
- **CHARINDEX('string to be searched', col)**: Return the numeric position of a named character.
- **REPLACE(string, from, to)**: Searches a text expression for a character string and, if found, replaces it with a specified replacement string.
- **SUBSTRING(string,x,n)**: Is used to extract a specific part of a string.

DATE FUNCTIONS

Operate on values of date data.

- **GETDATE()**: Returns the current date and time.
- **DATEADD()**: Adds month, day or year to specified date.
- **DATENAME()**: returns the name of the specified date or time part (e.g., month name, day name).It provides a more human-readable output.
- **DATEPART()**:return the integer value of a specified part of a date. It returns an integer representing the specified date or time.
- **DATEDIFF()**: difference between the dates.
- **FORMAT(date, format)**: Converts a date or timestamp to a specified format.

EXAMPLES

```
select
DATEADD(month, 2, '2-19-2025') AS twomonthsahead,
DATEADD(DAY, 2, '2-19-2025') as twodaysahead,
DATEADD(YEAR, -2, '2-19-2025')as twoyearsago;
```

twomonthsahead	twodaysahead	twoyearsago
2025-04-18 00:00:00.000	2025-02-20 00:00:00.000	2023-02-18 00:00:00.000

```
select
DATENAME(MONTH, '2-19-2025') AS CurrentMonth,
DATENAME(DAY, '2-19-2025') AS CurrentDate,
DATENAME(WEEKDAY, '2-19-2025') AS CurrentDay,
DATENAME(YEAR, '2-19-2025') AS Currentyear;
```

CurrentMonth	CurrentDate	CurrentDay	Currentyear
February	18	Tuesday	2025

```
select
DATEPART(MONTH, '2-19-2025') AS CurrentMonth,
DATEPART(DAY, '2-19-2025') AS CurrentDate,
DATEPART(WEEKDAY, '2-19-2025') AS CurrentDay,
DATEPART(YEAR, '2-19-2025') AS Currentyear;
```

CurrentMonth	CurrentDate	CurrentDay	Currentyear
2	18	3	2025

EXAMPLES

SELECT

```
DATEDIFF(HOUR, '2-18-2025', GETDATE()) as hours,  
DATEDIFF(MINUTE, '2-18-2025', GETDATE()) as minutes,  
DATEDIFF(SECOND, '2-18-2025', GETDATE()) as seconds;
```

```
select empno, ename,  
datediff(year, HIREDATE, GETDATE())  
as yearstillnow from emp;
```

hours	minutes	seconds
0	45	2705

	empno	ename	yearstillnow
1	7369	HAIDER	41
2	7499	AWAIS	41
3	7521	SHEHROZ	40
4	7566	KHALID	41
5	7654	KATERINA	41
6	7698	SAFDAR	40
7	7782	SHUJAAT	40
8	7788	FAIZAN	40
9	7839	BASHIR	40
10	7844	RUDABA	40
11	7876	MOHSIN	40
12	7900	NELSON	40
13	7902	LUBNA	41
14	7934	ADNAN	41
15	7935	M ALI	24

EXAMPLES

```
SELECT empno, FORMAT(hiredate, 'dd-MMM-yyyy') AS  
formatted_hire_date FROM emp;
```

	empno	formatted_hire_date
1	7369	13-Jun-1983
2	7499	15-Aug-1983
3	7521	26-Mar-1984
4	7566	31-Oct-1983
5	7654	05-Dec-1983
6	7698	11-Jun-1984
7	7782	14-May-1984
8	7788	05-Mar-1984
9	7839	09-Jul-1984
10	7844	04-Jun-1984
11	7876	04-Jun-1984
12	7900	23-Jul-1984
13	7902	05-Dec-1983
14	7934	21-Nov-1983
15	7935	01-Sep-2000

CONVERSION FUNCTIONS

Conversion functions in SQL are used to convert one data type to another. These functions help ensure compatibility between different data types within SQL queries.

1. **CAST(x):**

2. **CONVERT():** (especially for dates).

both can be used for data type conversion and formatting

Example:

```
SELECT stdName,  
CONVERT(VARCHAR, StdDOB, 101)  
AS formatted_date FROM std;
```

	stdName	formatted_date
1	Bisma	02/02/2000
2	Sadia	01/01/2001
3	Sarah	03/05/2001
4	Namra	06/19/2001
5	Waqar	09/10/1999
6	Noman	08/25/2001
7	Abrar	01/05/2001
8	Mohsin	03/23/2001
9	Sikandar	09/13/2001
10	Javed	10/21/2001

```

SELECT stdName,
CONVERT(VARCHAR, StdDOB, 100) AS formatted_date1,
CONVERT(VARCHAR, StdDOB, 101) AS USA_STANDARD_Format,
CONVERT(VARCHAR, StdDOB, 102) AS formatted_date3,
CONVERT(VARCHAR, StdDOB, 103) AS British_French_format,
CONVERT(VARCHAR, StdDOB, 110) AS USA_FORMAT,
CONVERT(VARCHAR, StdDOB, 120) AS ODBC
FROM std;

```

	stdName	formatted_date1	USA_STANDARD_Format	formatted_date3	British_French_format	USA_FORMAT	ODBC
1	Bisma	Feb 2 2000 12:00AM	02/02/2000	2000.02.02	02/02/2000	02-02-2000	2000-02-02 00:00:00
2	Sadia	Jan 1 2001 12:00AM	01/01/2001	2001.01.01	01/01/2001	01-01-2001	2001-01-01 00:00:00
3	Sarah	Mar 5 2001 12:00AM	03/05/2001	2001.03.05	05/03/2001	03-05-2001	2001-03-05 00:00:00
4	Namra	Jun 19 2001 12:00AM	06/19/2001	2001.06.19	19/06/2001	06-19-2001	2001-06-19 00:00:00
5	Waqar	Sep 10 1999 12:00AM	09/10/1999	1999.09.10	10/09/1999	09-10-1999	1999-09-10 00:00:00
6	Noman	Aug 25 2001 12:00AM	08/25/2001	2001.08.25	25/08/2001	08-25-2001	2001-08-25 00:00:00
7	Abrar	Jan 5 2001 12:00AM	01/05/2001	2001.01.05	05/01/2001	01-05-2001	2001-01-05 00:00:00
8	Mohsin	Mar 23 2001 12:00AM	03/23/2001	2001.03.23	23/03/2001	03-23-2001	2001-03-23 00:00:00
9	Sikandar	Sep 13 2001 12:00AM	09/13/2001	2001.09.13	13/09/2001	09-13-2001	2001-09-13 00:00:00
10	Javed	Oct 21 2001 12:00AM	10/21/2001	2001.10.21	21/10/2001	10-21-2001	2001-10-21 00:00:00

GENERAL/CONDITIONAL FUNCTIONS

These functions perform various operations on data and are often used for data manipulation, conversion, and control flow.

1. **COALESCE ()**: return the first non-null expression. (It's particularly useful when dealing with nullable columns or situations where you want to provide a default value if a column has a null value.)

EXAMPLE:

```
SELECT stdName,  
COALESCE(stdaddress, 'NotAvailable')  
AS updatedContact FROM std;
```

stdName	updatedContact
Bisma	Not Available
Sadia	Not Available
Sarah	Not Available
Namra	Not Available
Waqar	Not Available
Noman	Not Available
Abrar	Not Available
Mohsin	Not Available
Sikandar	Not Available
Javed	Not Available

2. **ISNULL()**: Is used to replace NULL values with a specified replacement value. It takes only 2 arguments.

```
SELECT stdname, ISNULL(stdemail, Stdphone) FROM std;
```

3. **CASE()**: conditional expression, similar to the **IF-THEN-ELSE** structure.

CASE

WHEN condition1 THEN result1

WHEN condition2 THEN result2

ELSE defaultResult

END

EXAMPLES

```
select ename, job, sal,  
case
```

```
when sal between 500 and 1500 then 'low salary'
```

```
when sal between 1500 and 3000 then 'average salary'
```

```
when sal between 3000 and 4000 then 'above average salary'
```

```
when sal > 4000 then 'high salary'
```

```
else 'lowest salary'
```

```
end as Condition from emp;
```

	ename	job	sal	Condition
1	HAIDER	CLERK	800.00	low salary
2	AWAIS	SALESMAN	1600.00	average salary
3	SHEHROZ	SALESMAN	1250.00	low salary
4	KHALID	MANAGER	2975.00	average salary
5	KATERINA	SALESMAN	1250.00	low salary
6	SAFDAR	MANAGER	2850.00	average salary
7	SHUJAAT	MANAGER	2450.00	average salary
8	FAIZAN	ANALYST	3000.00	average salary
9	BASHIR	PRESIDENT	5000.00	high salary
10	RUDABA	SALESMAN	1500.00	low salary
11	MOHSIN	CLERK	1100.00	low salary
12	NELSON	CLERK	950.00	low salary
13	LUBNA	ANALYST	3000.00	average salary
14	ADNAN	CLERK	1300.00	low salary
15	M ALI	NETWORK...	4000.00	above avera...

Select top 5 employee_id, first_name, last_name
from employees
order by first_name

	employee_id	first_name	last_name
	121	Adam	Fripp
	103	Alexander	Hunold
	115	Alexander	Khoo
	193	Britney	Everett
	104	Bruce	Ernst
	179	Charles	Johnson
	109	Daniel	Faviet
	105	David	Austin
	114	Den	Raphaely



	employee_id	first_name	last_name
▶	121	Adam	Fripp
	115	Alexander	Khoo
	103	Alexander	Hunold
	193	Britney	Everett
	104	Bruce	Ernst